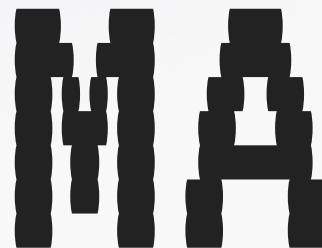
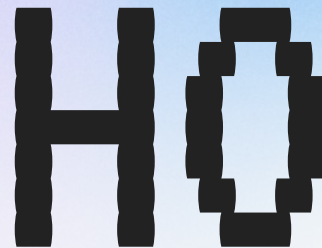
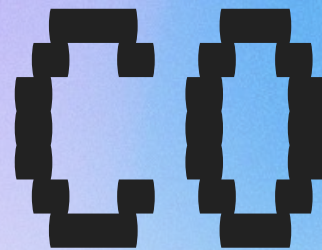


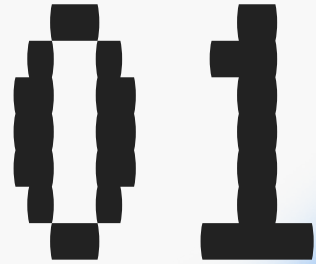
# ROVER BRUSHLESS MOTOR ADAPTATION



**Halfway Presentation**

Francesco Bighiani | Chahid El Khamlichi | Hugo Mériaux

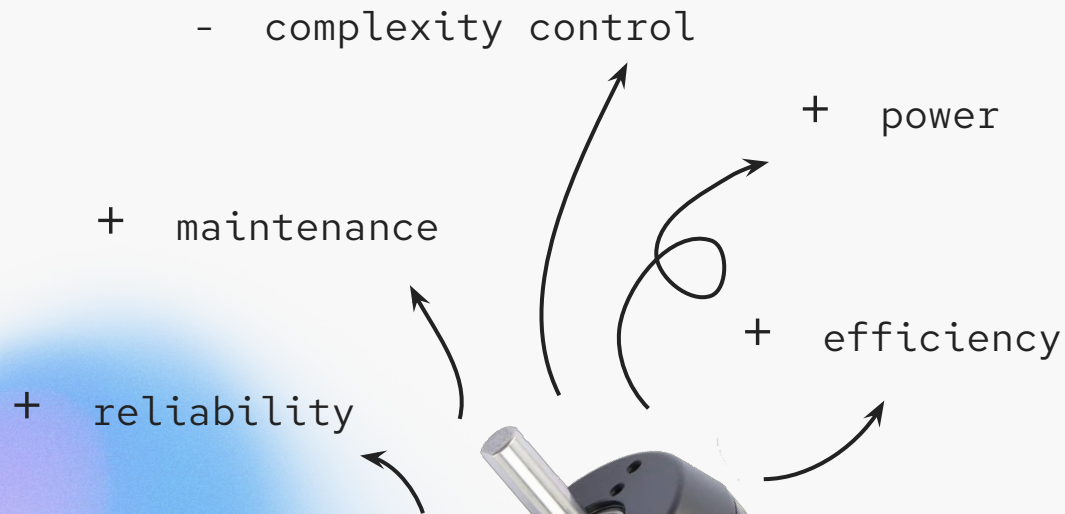
- 01 Overview of the project
- 02 Getting started with the ESC
- 03 Engine control software
- 04 Mechanical adaptation of brushless motors
- 05 Future progress



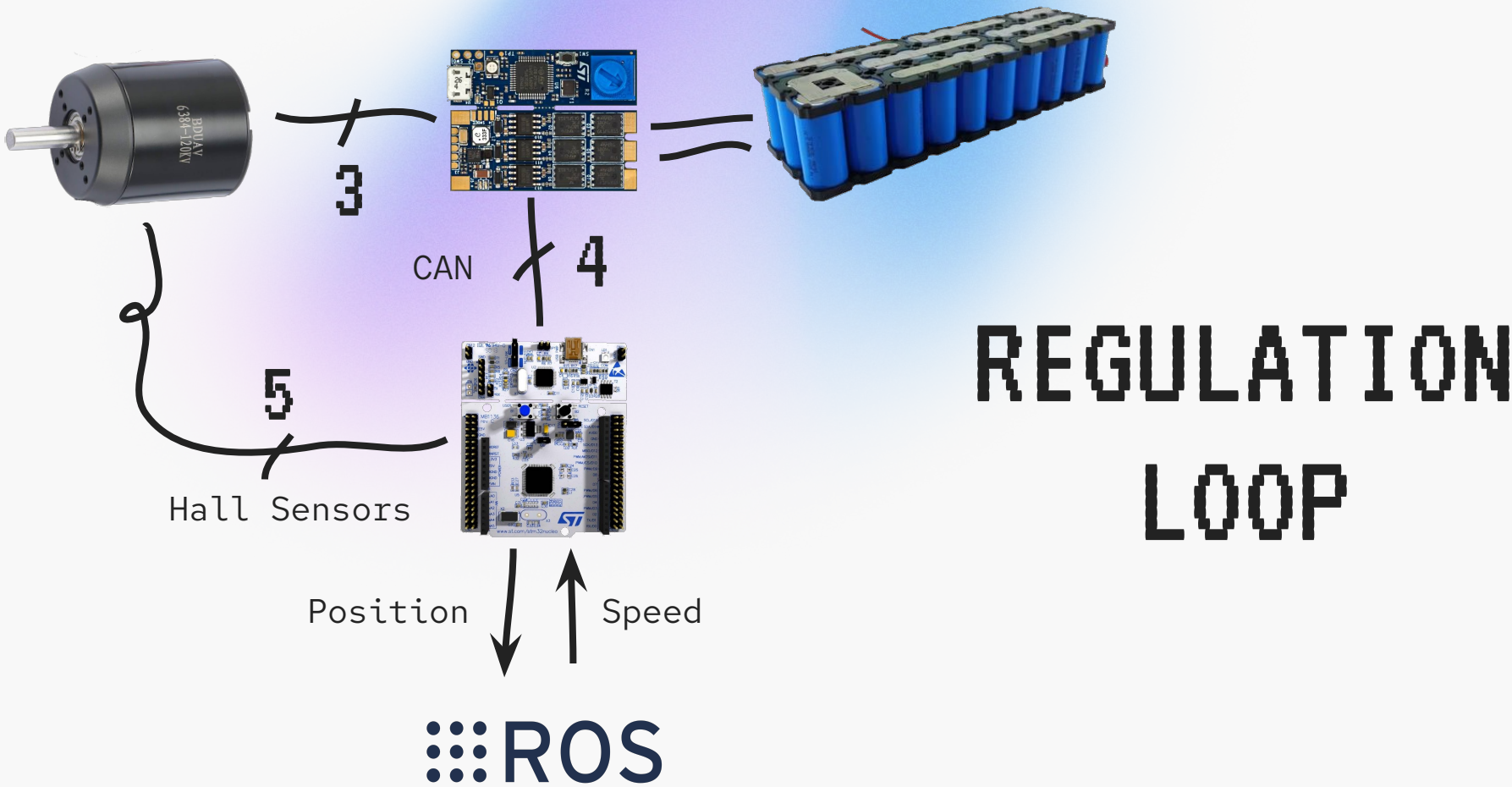
# Overview of the project

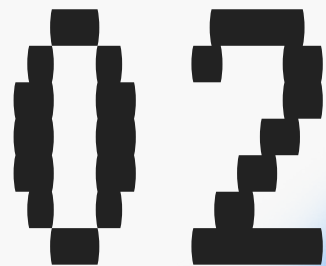


**DC MOTOR**



**BRUSHLESS MOTOR**





Getting started  
with the ESC

# B-G431B-ESC1 SPECS

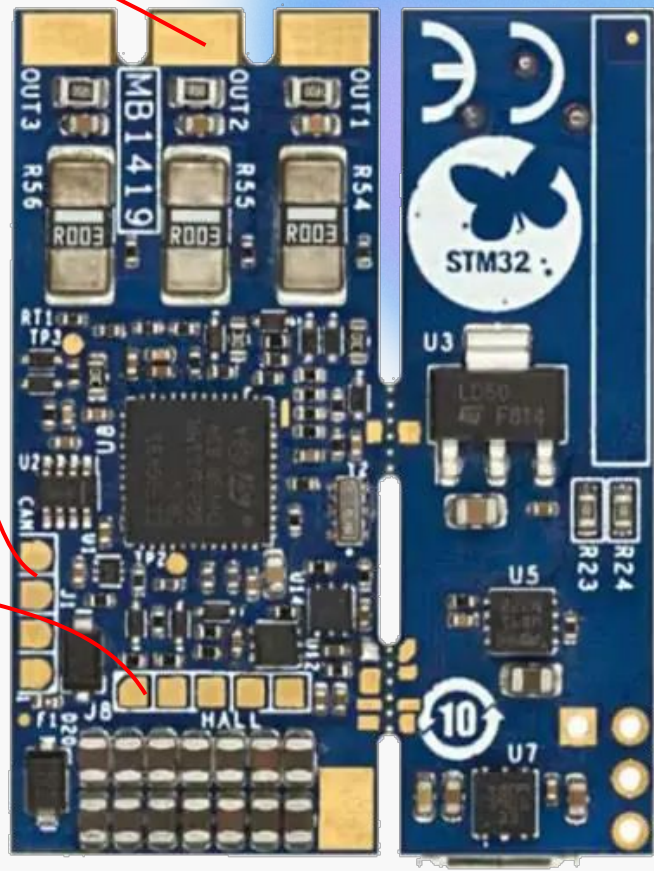


Output (triphas)  
40 A

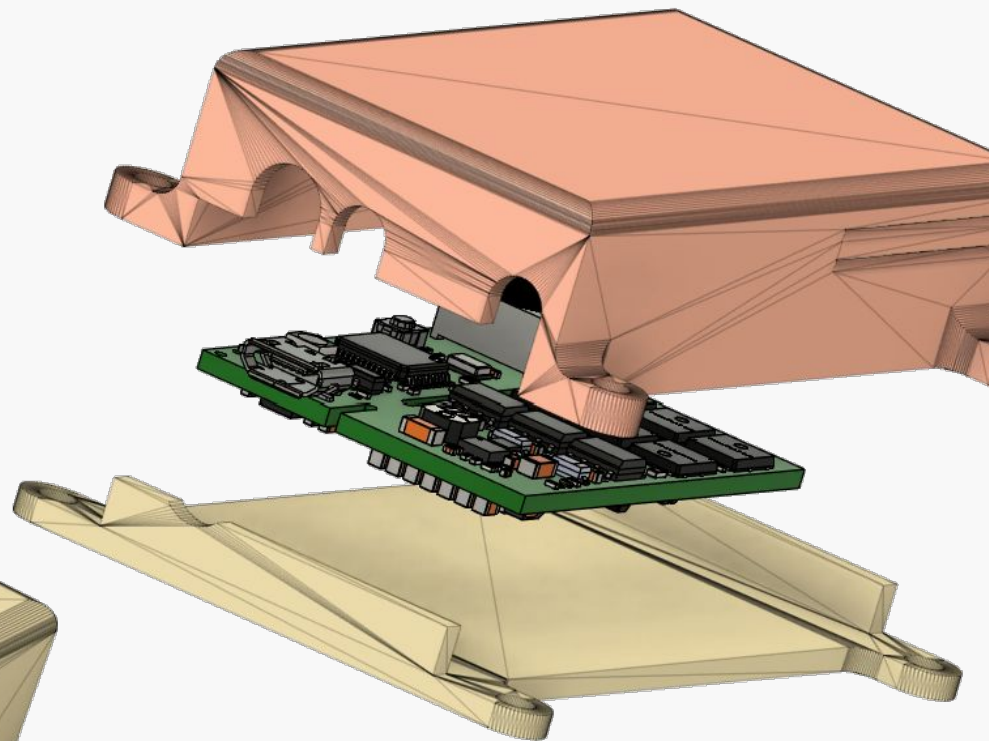
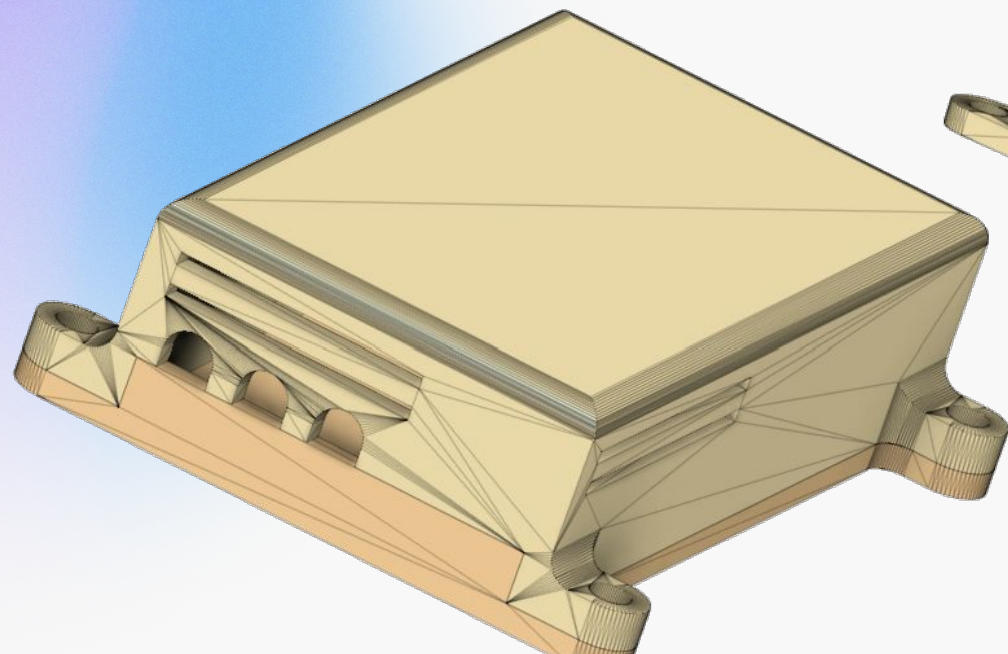
CAN

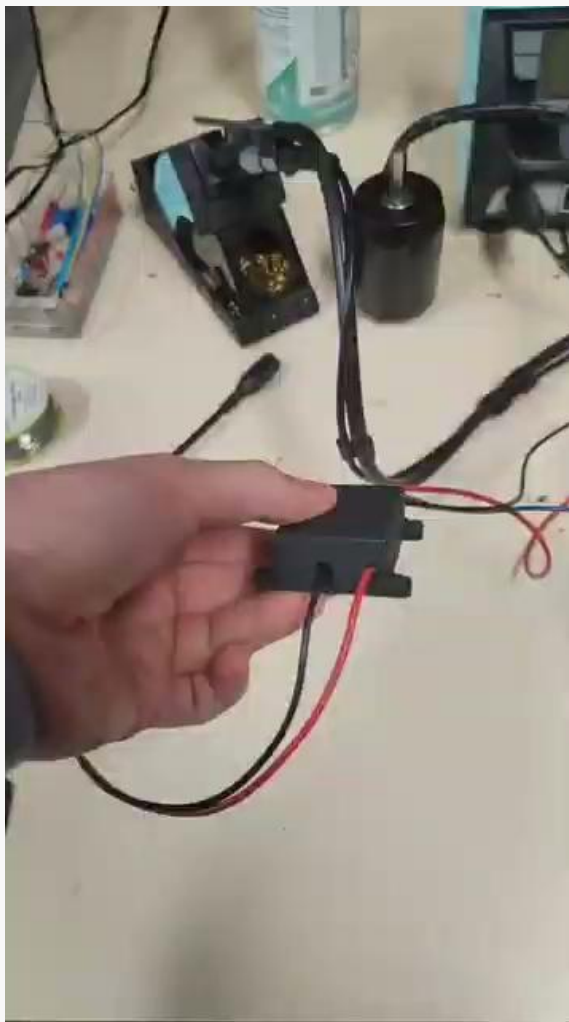
HALL sensor

Input | 13V to 24V



# B-G431B-ESC1 BOX







Engine control  
software

# From Drone ESC to Rover Actuator

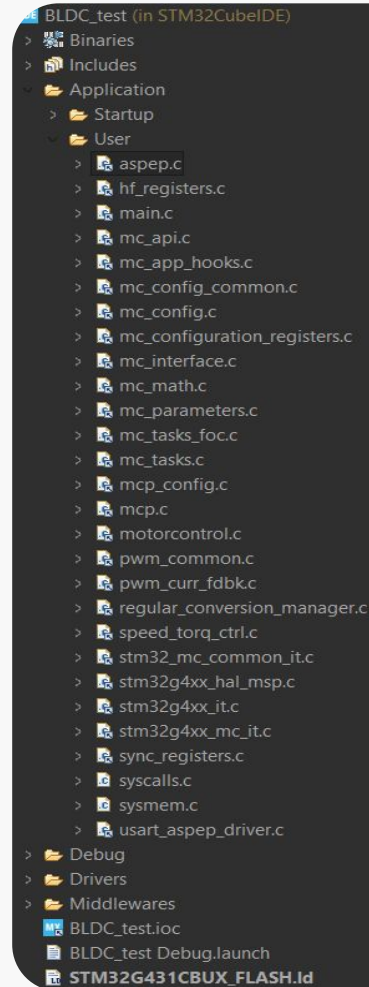
## Material & Equipment:

### Hardware :

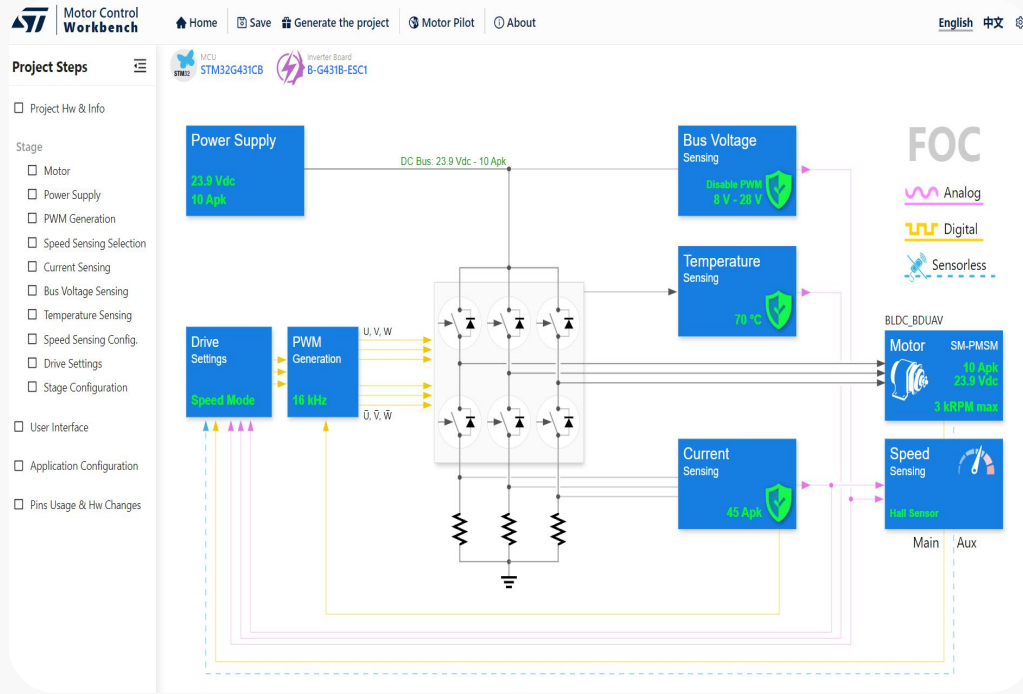
- B-G431B-ESC1 (Controller)
- ROXXY C28-27-26 (Test Motor)
- BDUAV 6384-120KV (Rover Motor)

### Software :

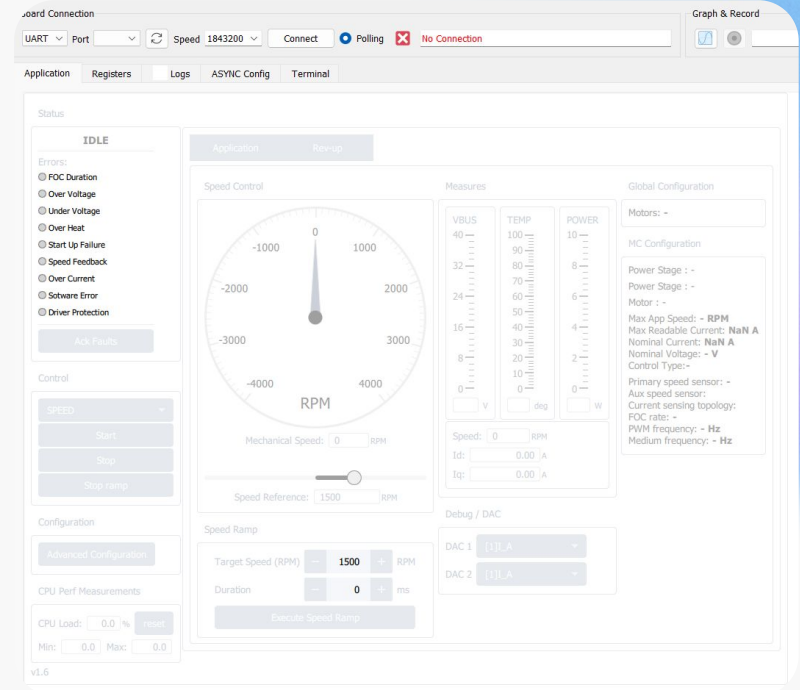
- STM32CubeMX
- STM32CubeIDE
- MotorPilot
- MotorControl Workbench



# Software Used :

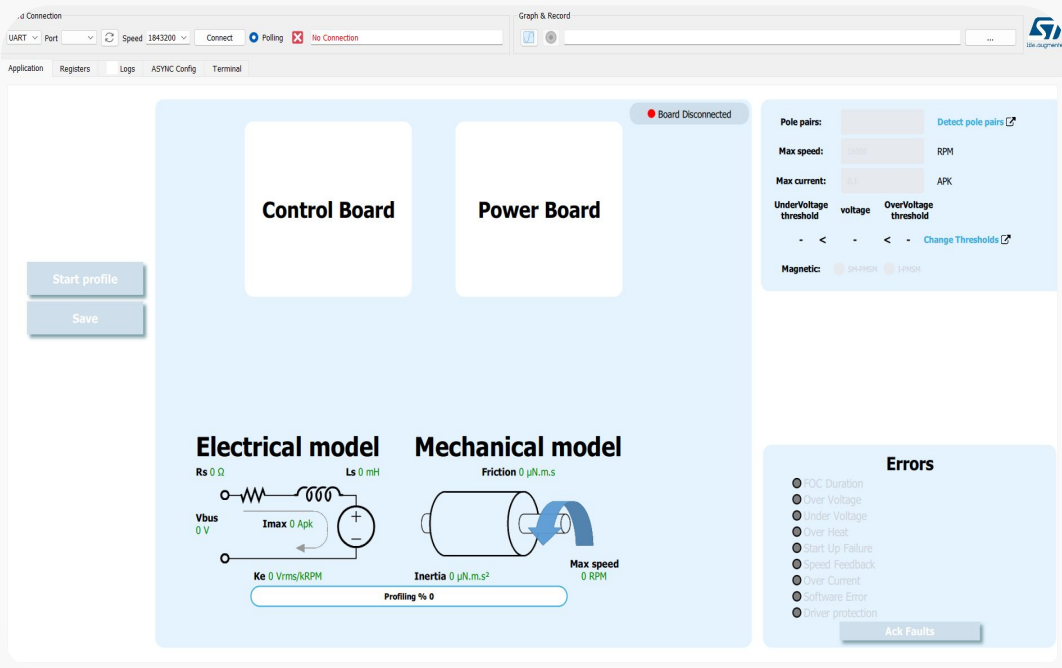


MC Workbench

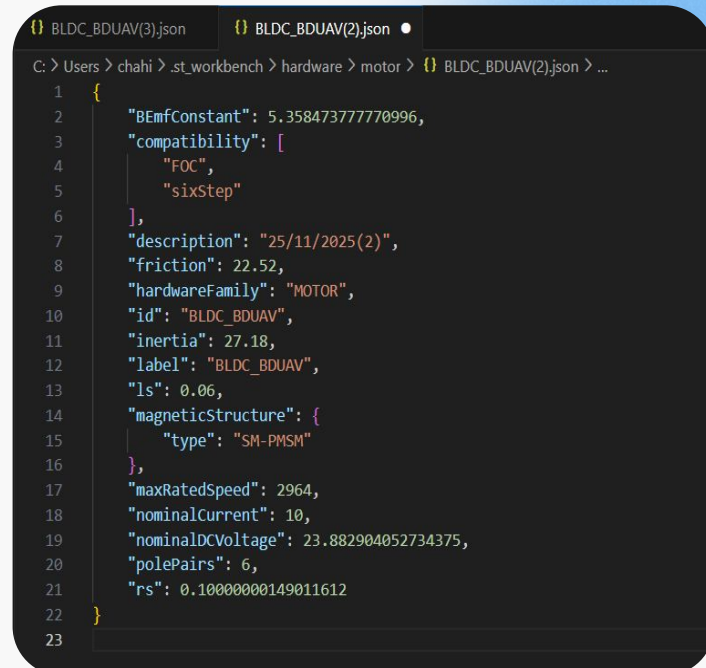


Motor Pilot

# Software Used :



Motor profiler



Results

# Reverse Engineering Communication & Solving Protection Issues

USB-Control (1st Week), Protocol Analysis (Week 2 and 3), Fault Analysis (Week 4)

Stable Control (Week 5). Terminal & Label from MotorPilot

ST Motor Pilot V6.4.1 - MC\_FOC\_SDK  
GUI Configuration Help Language

Board Connection  
UART Port Speed 1843200 Connect Polling No Connection

Graph & Record

Application Registers Logs ASYNC Config Terminal

| Time        | Data                  | Human Reading                                       |
|-------------|-----------------------|-----------------------------------------------------|
| 17:32:54... | → Tx 8904000011001... | Read Register ...                                   |
| 17:32:54... | ← Rx E97F00C0F77...   | ASPEP_ASYNC_RX ASYNC RX                             |
| 17:32:54... | ← Rx 0A05003004000... | ASPEP_RESP MCP Cmd OK                               |
| 17:32:54... | ← Rx E97F00C0F97B...  | ASPEP_ASYNC_RX ASYNC RX                             |
| 17:32:54... | ← Rx E97F00C0F57F...  | ASPEP_ASYNC_RX ASYNC RX                             |
| 17:32:54... | → Tx 8904000011001... | Read Register ...                                   |
| 17:32:54... | ← Rx E97F00C0F183...  | ASPEP_ASYNC_RX ASYNC RX                             |
| 17:32:54... | ← Rx 0A05003004000... | ASPEP_RESP MCP Cmd OK                               |
| 17:32:55... | ← Rx E97F00C0ED87...  | ASPEP_ASYNC_RX ASYNC RX                             |
| 17:32:55... | ← Rx E97F00C0E98B...  | ASPEP_ASYNC_RX ASYNC RX                             |
| 17:32:55... | → Tx 8904000011001... | Read Register ...                                   |
| 17:32:55... | ← Rx E97F00C0E58F...  | ASPEP_ASYNC_RX ASYNC RX                             |
| 17:32:55... | ← Rx 0A05003004000... | ASPEP_RESP MCP Cmd OK                               |
| 17:32:55... | ← Rx E97F00C0E193...  | ASPEP_ASYNC_RX ASYNC RX                             |
| 17:32:55... | ← Rx E97F00C0D097...  | ASPEP_ASYNC_RX ASYNC RX                             |
| 17:32:55... | → Tx D90000400800...  | Write Register ASYNC_UARTA 0x2905070000080000FF0000 |
| 17:32:55... | ← Rx E97F00C0D99B...  | ASPEP_ASYNC_RX ASYNC RX                             |
| 17:32:55... | ← Rx 1A00002000       | ASPEP_RESP MCP Cmd OK                               |
| 17:32:55... | → Tx 8904000011001... | Read Register ...                                   |
| 17:32:55... | ← Rx 890900D0D59F...  | ASPEP_ASYNC_RX ASYNC RX                             |
| 17:32:55... | ← Rx 0A05003004000... | ASPEP_RESP MCP Cmd OK                               |
| 17:32:55... | → Tx 8904000011001... | Read Register ...                                   |
| 17:32:55... | ← Rx 0A05003004000... | ASPEP_RESP MCP Cmd OK                               |
| 17:32:56... | → Tx 8904000011001... | Read Register ...                                   |
| 17:32:56... | ← Rx 0A05003004000... | ASPEP_RESP MCP Cmd OK                               |
| 17:32:56... | → Tx 8904000011001... | Read Register ...                                   |
| 17:32:56... | ← Rx 0A05003004000... | ASPEP_RESP MCP Cmd OK                               |
| 17:32:56... | → Tx 8904000011001... | Read Register ...                                   |
| 17:32:56... | ← Rx 0A05003004000... | ASPEP_RESP MCP Cmd OK                               |
| 17:32:56... | → Tx 8904000011001... | Read Register ...                                   |

Label

Value

▼ Frame ASPEP

ASPEP HEADER[0-32]

Type [0-3]

payloadlength [4-16]

reserved [17-27]

CRCH [29-31]

▼ Payload[32-...

Timestamp[32-63]

ASYNCRID[...]

Mark[...]

CRCH[...]

0xE97F00C0E193BD02C...

0xE97F00C0

ASPEP ASYNCR

2046

12

0xE193BD02CEFFEEFFD...

0xE193BD02 (45978593)

to do

to do

Not implemented yet [...]

Log Size 1002

CPU= 7.1%

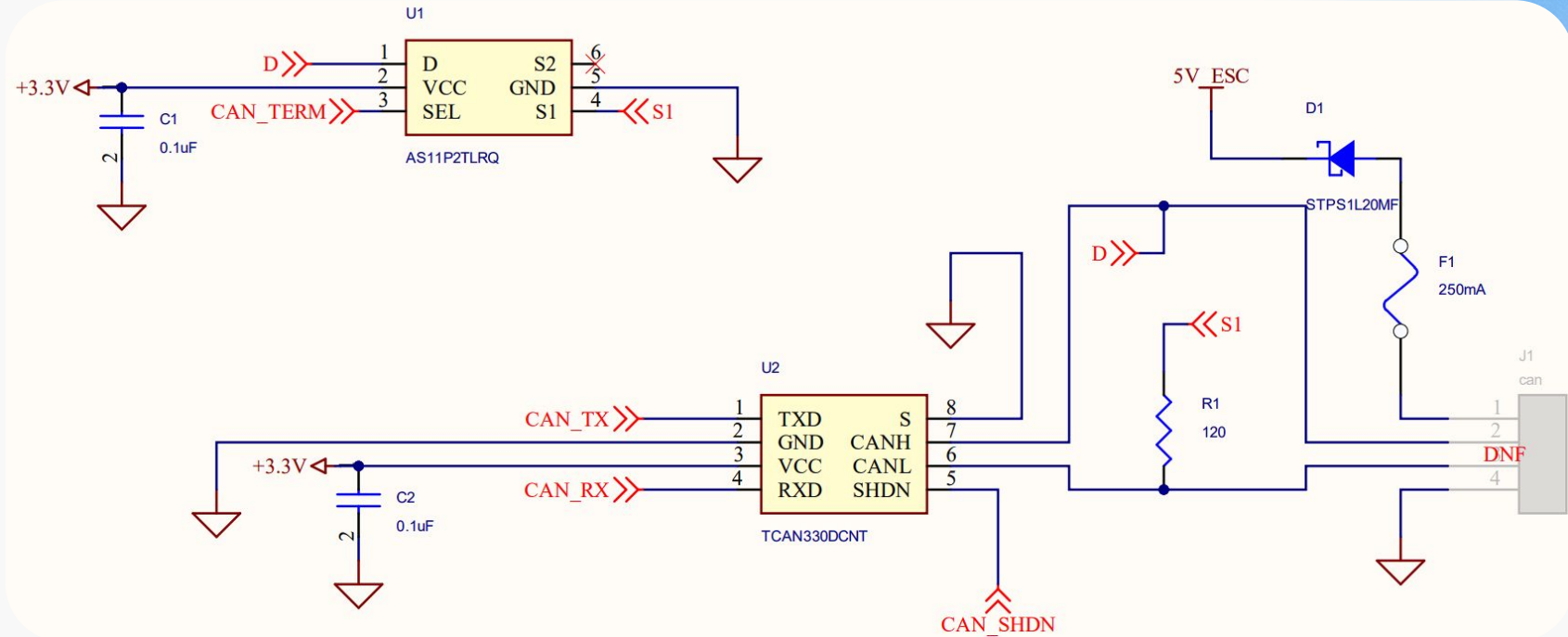
|                             |        |
|-----------------------------|--------|
| Reg 25[FAULTS_FLAGS]        | 0x0019 |
| Reg 89[SPEED_MEAS]          | 0x0059 |
| Reg 7001[MOTOR_POWER]       | 0x1B59 |
| Reg 153[SPEED_REF]          | 0x0099 |
| Reg 217[STOPLL_EST_BEMF]    | 0x00D9 |
| Reg 281[STOPLL_OBS_BEMF]    | 0x0119 |
| Reg 657[I_D_KP]             | 0x0291 |
| Reg 721[I_D_KI]             | 0x02D1 |
| Reg 2385[I_Q_REF]           | 0x0951 |
| Reg 401[I_Q_KP]             | 0x0191 |
| Reg 465[I_Q_KI]             | 0x01D1 |
| Reg 5265[SPEED_KI_DIV]      | 0x1491 |
| Reg 5201[SPEED_KP_DIV]      | 0x1451 |
| Reg 145[SPEED_KP]           | 0x0091 |
| Reg 209[SPEED_KI]           | 0x00D1 |
| Reg 2449[I_D_REF]           | 0x0991 |
| Reg 2257[I_Q_MEAS]          | 0x08D1 |
| Reg 2321[I_D_MEAS]          | 0x0911 |
| Reg 1489[HEATS_TEMP]        | 0x05D1 |
| Reg 1425[BUS_VOLTAGE]       | 0x0591 |
| Reg 849[STOPLL_C1]          | 0x0351 |
| Reg 913[STOPLL_C2]          | 0x0391 |
| Reg 1169[STOPLL_KP]         | 0x0491 |
| Reg 1105[STOPLL_KI]         | 0x0451 |
| Reg 6417[STOPLL_KP_DIV]     | 0x1911 |
| Reg 6353[STOPLL_KI_DIV]     | 0x18D1 |
| Reg 2961[STOPLL_ROT_SPEED]  | 0x0B91 |
| Reg 2897[STOPLL_EL_ANGLE]   | 0x0B51 |
| Reg 3025[STOPLL_I_ALPHA]    | 0x0BD1 |
| Reg 3089[STOPLL_I_BETA]     | 0x0C11 |
| Reg 3153[STOPLL_BEMF_ALPHA] | 0x0C51 |
| Reg 3217[STOPLL_BEMF_BETA]  | 0x0C91 |
| Reg 73[STATUS]              | 0x0049 |
| Reg 137[CONTROL_MODE]       | 0x0089 |
| Reg 201[RUC_STAGE_NBR]      | 0x00C9 |

CRC

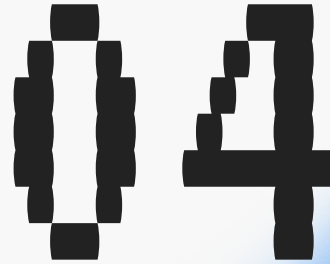
Not impl

# Building a Robust Industrial Communication Layer

Confirmed CAN transceiver (TCAN330D) and 120 $\Omega$  termination circuit







Mechanical  
adaptation of  
brushless motors

# BDUAV 6384-120KV SPECS



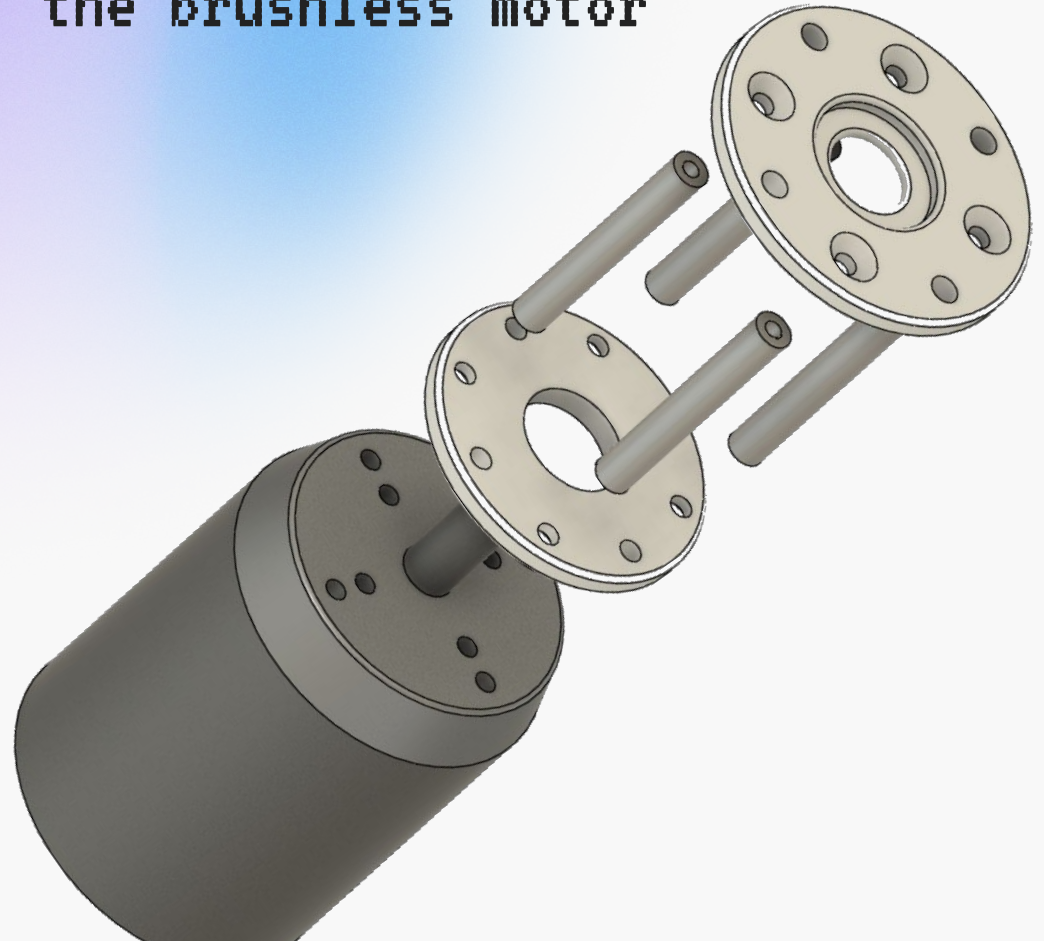
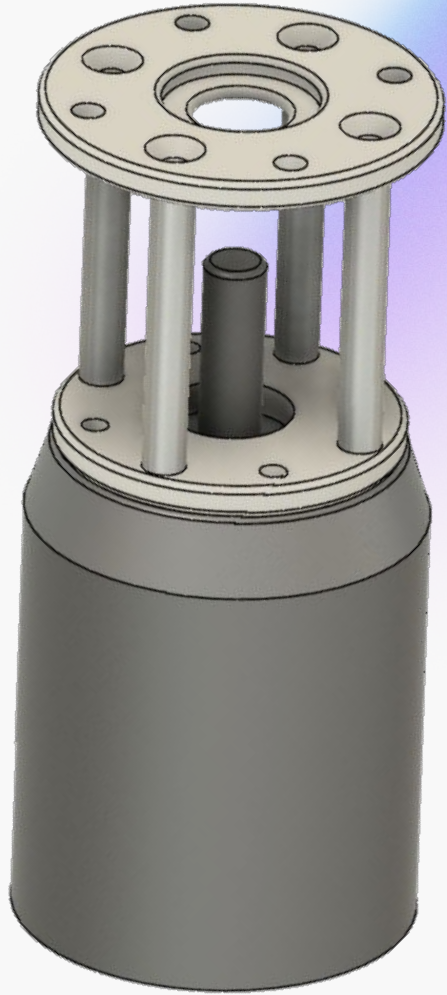
Idling max speed :  $120 \times 25 =$   
3000 RPM

Real max speed (measured) :  
2964 RPM

Input 24V to 48V  
(more than 100A depending on  
Load)

2970 W

Adaptation from the gearbox to  
the brushless motor



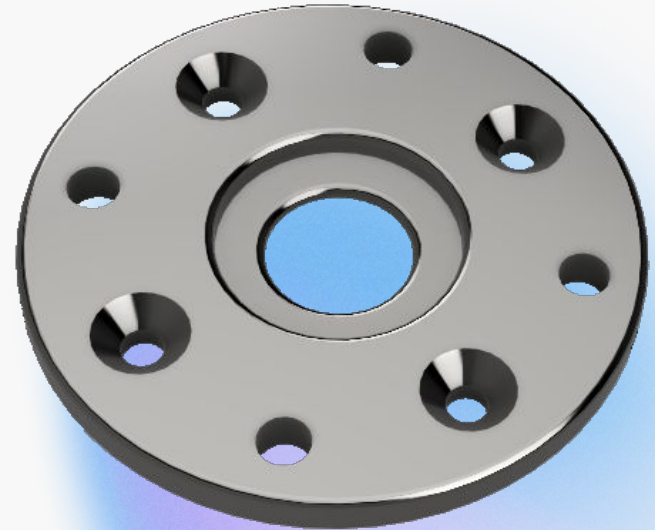


05

Future progress

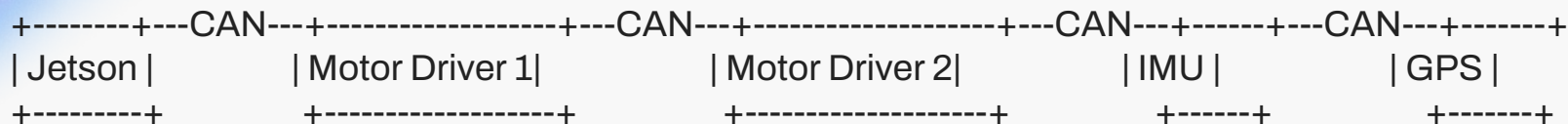
# Machined aluminum parts for the motor adaptor:

- Stronger, more durable in the field
- Rigid, heat-resistant; won't deform under load
- Improves alignment; reduces noise
- Test part ordered (ENSEA workshop)



# Implementing CAN connectors for communication between robot components

- Reliable in the field (less noise issues than micro-USB)
- One shared bus for multiple boards in parallel (each board has its own ID)
- Locking connector reduces accidental disconnects



# Conclusions

- Defined and validated the overall architecture for the brushless actuation upgrade (control, power, and sensing)
- Learned and validated the motor control workflow on the ESC (configuration, commands, safety limits)
- Tested open-loop motor operation at 24 V using current-based control (Hall sensor feedback not connected yet)
- Built and tested the mechanical integration concept, identifying improvements for the final machined adapter

THA

Any Questions?

NHS